

PROGRAM {BEST FIT}

```
#include <stdio.h>
#include <stdlib.h>

#define MAX 50

struct BLOCK {
    int size;
    int isallocated;
    int pallocated;
    int psize;
    int frag;
} block[MAX];

int main() {
    int nb, np, p_size[MAX];

    printf("Enter the no: of blocks\n");
    scanf("%d", &nb);

    printf("Enter the size of the blocks\n");
    for (int i = 0; i < nb; i++) {
        printf("Block%d: ", i);
        scanf("%d", &block[i].size);
        if (block[i].size < 0) {
            printf("Invalid Input\n");
            exit(0);
        }
        block[i].isallocated = 0;
        block[i].pallocated = -1;
        block[i].psize = 0;
        block[i].frag = block[i].size;
    }

    printf("Enter the no: of processes\n");
    scanf("%d", &np);

    printf("Enter the size of the processes\n");
    for (int i = 0; i < np; i++) {
        printf("Process%d: ", i);
        scanf("%d", &p_size[i]);
        if (p_size[i] < 0) {
            printf("Invalid Input\n");
            exit(0);
        }
    }
}

// Allocation logic: Best Fit
for (int i = 0; i < np; i++) {
    int best_idx = -1;
    for (int j = 0; j < nb; j++) {
        if (block[j].isallocated == 0 && block[j].size >= p_size[i]) {
            if (best_idx == -1 || block[j].size < block[best_idx].size) {
                best_idx = j;
            }
        }
    }
}
```

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    }

    if (best_idx != -1) {
        block[best_idx].isallocated = 1;
        block[best_idx].pallocated = i;
        block[best_idx].psize = p_size[i];
        block[best_idx].frag = block[best_idx].size - p_size[i];
        printf("Process%d is allocated to block%d\n", i, best_idx);
    }
}

// Display Output
printf("\nBlock\tIsallocated\tProcess_allocated\tBlockSize\tPSize\tFragment\n");
for (int i = 0; i < nb; i++) {
    printf("%d\t%d\t\t", i, block[i].isallocated);
    if (block[i].isallocated == 1)
        printf("%d\t\t\t", block[i].pallocated);
    else
        printf("-\t\t\t");

    printf("%d\t\t%d\t%d\n", block[i].size, block[i].psize, block[i].frag);
}

return 0;
}

```

OUTPUT {BEST FIT}

blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_8\$ gcc bestfit.c

blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_8\$./a.out

Enter the no: of blocks

4

Enter the size of the blocks

Block0: 10

Block1: 20

Block2: 30

Block3: 40

Enter the no: of processes

5

Enter the size of the processes

Process0: 12

Process1: 22

Process2: 35

Process3: 8

Process4: 10

Process0 is allocated to block1

Process1 is allocated to block2

Process2 is allocated to block3

Process3 is allocated to block0

Block	Isallocated	Process_allocated	BlockSize	PSize	Fragment
0	1	3	10	8	2
1	1	0	20	12	8
2	1	1	30	22	8
3	1	2	40	35	5

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